Quiz 2

Graded

Student

Clarisse Cheung

Total Points

0 / 5 pts

Question 1

Question 1

0 / 1.5 pts

- **– 0 pts** Full Credit
- **– 0.15 pts** Majority Credit
- **– 0.75 pts** Half Credit
- ✔ **– 1.5 pts** No Credit
- **– 0 pts** Found $\ell = 2$ or $\ell = 3$, but not both
- **– 0 pts** No answer given
- **– 0 pts** Did not find $\ell = 2$ or $\ell = 3$
- **– 0 pts** Did not make any claim about $\ell \geq 4$
- **– 0 pts** Incomplete justification
- ✔ **– 0 pts** Incorrect justification

  3 How does the attacker use this fact to break the cipher?

- **– 0 pts** Confusing justification

Question 2

Question 2

0 / 1.5 pts

- **– 0 pts** Full Credit
- **– 0.15 pts** Majority Credit
- **– 0.75 pts** Half Credit
- ✔ **– 1.5 pts** No Credit
- **– 0 pts** Correct proof, missing/incorrect logic for $\Pr\left[\mathrm{Enc}_k(m) = c\right] = \frac{1}{2^\ell}$.
- ✔ **– 0 pts** Attempted to solve via perfectly secrecy rather than perfect indistinguishability.

2 I recommend solving this by showing for all $m \in M$, $c \in C$ there exists a unique $k \in K$ such that $\mathrm{Enc}_k(m) = m \oplus k = c$. Once you have this, you can prove the statement via perfect indistinguishability.

Question 3

0 / 2 pts

– **0 pts** Full Credit

– **0.2 pts** Majority Credit

– **1 pt** Half Credit

✔ – **2 pts** No Credit

1 This is not a proof. I recommend picking a message-ciphertext pair (m, c) such that the probability that m encrypts to c is zero. Then, find a different m' that has a nonzero probability of encrypting to c. This would contradict perfect indistinguishability.

CMSC 28400 **Quiz 2 (3 questions)** October 6, 2026

Name: Clarisse Cheung **CNET ID:** clarissec@uchicago.edu

Please write neatly, not too small, and not too faint.

Question 1: Assume an attacker knows that Alice encrypted one of two messages: $m_0 =$ ABCD or $m_1 =$ EFGZ. Suppose the attacker learns that c is an encryption of the message using Vigenère cipher with key length $\ell \geq 2$. For what values of ℓ can the attacker determine the message with certainty? Justify your answer.

ignore

Since the Vigenère cipher uses a key of length $\ell \geq 2$, and the messages are of length 4, the attacker can try key lengths of 2, 3, and 4. Using the formula $[i_3 - 1] \bmod \ell + 1$ with i being the jth character of either m_0 or m_1, the attacker can find the position

Since m_0 and m_1 are of length 4, any $l \leq 4$ and $l \geq 2$ as given, are allowing the attacker to find the message. Since the cipher repeats the key when $|m| > \ell$.

3

Question 2: Prove the following: For every $\ell > 0$, the ℓ-bit one-time pad scheme OTP_ℓ is perfectly secret.

For OTP_ℓ, the scheme is $(m \oplus k, k \oplus c)$. Since each bit in the key of length ℓ is random and independent, it generates a random ciphertext c. $Pr[M=m \mid Enc_k(m)=c] = Pr[M=m]$ or $Pr[Enc_k(m)=c] = 0$. Since c is random, $Pr[M=m \mid Enc_k(m)=c] = Pr[M=m]$.

$m \oplus k = c$
$k \oplus c = m$

2

Question 3: Fix $\ell > 0$. Let $\mathcal{M} = \mathcal{C} = \mathcal{K} = \{0,1\}^\ell$. Let OTP_ℓ be the ℓ-bit one-time pad encryption scheme.

We define a new encryption scheme as follows. Let $\mathcal{M}' = \mathcal{M} \setminus \{0^\ell\}$, $\mathcal{C}' = \mathcal{C}$, and $\mathcal{K}' = \mathcal{K} \setminus \{0^\ell\}$. Define the encryption scheme $\mathsf{OTP}'_\ell = \mathsf{OTP}_\ell$ (i.e., identical encryption and decryption algorithms) for message space $\mathcal{M}'$, ciphertext space $\mathcal{C}'$, and key space $\mathcal{K}'$.

Prove that OTP'_ℓ is not perfectly secret.

Take $m' \in M'$, $m' = 111$. $k' \in K'$ and $k' = 111$. Encryption gives $m' \oplus k' = c' = 000$. Thus because c' is always 000, the attacker can guess $c \oplus k = 111$ since